JADE OS · Benefits & Features Technical Brief

Prepared by JADE OS · Operations Workbench · June 07, 2026

This document is operator-grade. Every number is benchmark-traceable; no synthetic data appears.

1 · Executive Overview

JADE OS is the operator-grade AI layer that composes six freight-domain agents into a single, auditable workflow. This brief documents the platform's modules, data pipeline, decision logic, ROI rationale, implementation cadence, and the Minnesota mid-market opportunity in one document.

• Six agent modules · Dispatch · Route+Fuel · Compliance · Pricing · Retention · Maintenance

• L1/L2 autonomy levels by module · suggest-vs-act gating tuned to risk

• 90-day pilot template · 4-phase rollout · success metrics declared pre-pilot

• Rate-floor guard + immutable audit trail at the platform level, not per module

• Built on the same workflow-memory + claims/risk substrate you can audit in the console

2 · Module M1 · Dispatch Optimizer · Deep Dive

M1 consumes ELD HOS state, TMS open loads, driver preferences, and customer SLAs to score every candidate driver-load pairing in <50ms. The scoring blends cost-to-serve, deadhead minutes, HOS feasibility, and customer-priority weighting.

• Inputs · ELD HOS, TMS open loads (real-time), driver pref profile, lane rates

• Decision · constraint-satisfaction + utility (cost · HOS · SLA · driver fit)

• Action · recommend to dispatcher (L2) · on approval, writes to TMS via API

• KPIs · empty-mile reduction 8-15% · dispatcher load -30-50% · time-to-dispatch -40%

• Failure modes · stale TMS feed → falls back to last-known; driver-pref absent → uses fleet defaults

3 · Module M2 · Route + Fuel Agent · Deep Dive

M2 computes the cost-optimal route given current diesel pricing (EIA · regional), weather, DOT 511 incident feeds, and truck-restricted segments. It then plans fuel stops by solving a small MILP that picks the cheapest qualifying station within range that meets the next-leg dwell constraints.

• Inputs · O/D, EIA diesel, weather, DOT 511, truck-restricted layer, driver loyalty stops

• Decision · graph routing + fuel-arbitrage MILP

• Action · push optimized route + fuel-stop sequence to driver app

• KPIs · fuel savings 10-18% · OTD +6-12 pp · re-route count -50%

• Compliance · prefers driver-loyalty stops when within $0.05/gal of optimum

4 · Module M3 · Compliance + Safety · Deep Dive

M3 turns FMCSA Part 395 (HOS) and Part 396 (Vehicle Maintenance) into runnable rules. It runs every pre-trip and posts an audit-ready document set for every dispatch. Anomalies route to a human review queue.

• Inputs · ELD HOS, DQF, vehicle maint history, prior FMCSA inspections

• Decision · rules engine + anomaly detection on inspection history

• Action · pre-trip score 0-100; >70 hard-blocks dispatch and routes to safety

• KPIs · violation drop 40-60% · audit-pack prep · days → hours · CSA score trend

• Audit trail · every dispatch decision linked to compliance signature

5 · Module M4 · Pricing + Rate-Floor Guard · Deep Dive

M4 recommends sell rates using lane-rate history, spot/contract index, fuel surcharge, and capacity. Before any quote leaves the system, the Rate-Floor Guard validates against a per-lane / per-customer floor and emits a HARD block or SOFT warning. No quote bypasses this gate.

• Inputs · 24-mo lane rates, DAT/Greenscreens spot index, fuel surcharge, capacity availability

• Decision · elastic-net regression with rate-floor invariant

• Action · recommended sell rate + floor verdict (PASS / SOFT / HARD)

• KPIs · margin uplift 1.5-3.0 pp · below-floor quote attempts blocked · win-rate stable

• Hard block · only an authorized role can override, override is immutably audited

6 · Module M5 · Driver Lifecycle · Deep Dive

M5 scores driver retention risk continuously and surfaces personalized retention actions to dispatchers and driver-managers. It also drafts the next-best conversation using an LLM constrained by the driver's last 90 days of activity.

• Inputs · dispatch history, home-time requests, pay history, comm-channel engagement

• Decision · survival analysis + LLM coaching template (no fabrication; cites events)

• Action · weekly retention queue with recommended action per driver

• KPIs · 12-mo retention +6-12 pp · turnover cost saved $400-900 per driver · NPS lift

7 · Module M6 · Predictive Maintenance + Carrier Match · Deep Dive

M6 catches J1939 fault patterns and routes them to a recommended maintenance window. On the broker side, the same agent scores carriers for tendering — combining KPI history, claims, and recent volume.

• Inputs · J1939 fault codes, service history; (broker) carrier KPIs + claims + volume

• Decision · anomaly detection + reliability scoring

• Action · maintenance-window recommendation; carrier scorecard for tenders

• KPIs · unplanned downtime -20-30% · carrier on-time +5-10 pp

8 · ROI Analysis · Two Case Studies

Two illustrative archetypes drawn from public mid-market freight benchmarks (ATA · ATRI · BLS · FMCSA · EIA). All figures are operator-grade ranges, with sensitivity ±10%.

Case A · Small Regional · 50 trucks
• Annual savings ~ $380-460k · upfront $38k · license $60k
• Payback 6-8 months · 3-yr NPV ~ $720k @ 10% discount

Case B · Mid-Market · 175 trucks
• Annual savings ~ $2.0-2.4M · upfront $130k · license $210k
• Payback 2-3 months · 3-yr NPV ~ $4.5M @ 10% discount

Sources · ATA Trucking Industry Profile · ATRI Operational Costs of Trucking 2024 · BLS OEWS 53-3032 · EIA weekly diesel · FMCSA Safety Progress Report

9 · Implementation Roadmap · Four Phases

• Phase 1 · Readiness · 1 week · data inventory, integrations confirmed, success metrics signed

• Phase 2 · Baseline · 4 weeks · M1 + M3 in shadow mode, baseline measured

• Phase 3 · Pilot · 8-12 weeks · agents in L1/L2 production, weekly KPI review

• Phase 4 · Scale · ongoing · add M2/M4/M5/M6, quarterly business review, audit refresh

10 · Minnesota Opportunity & Technical Appendix

Minnesota's freight base — Twin Cities I-94/I-35 corridor, intermodal at Midway/CP terminals, agricultural outbound, and a deep mid-market fleet roster (Bay & Bay, Dart, Halvor Lines, ATS, Lakeville Motor Express, and dozens of 50-250 truck operators) — is a prime mid-market beachhead. Adoption tends to follow proof, so pilots are structured for fast metric capture.

Technical Appendix · APIs (POST /api/agent/dispatch/recommend, /route/optimize, /compliance/score, /pricing/quote, /retention/risk, /maintenance/window) · Auth model (tenant JWT, role-gated overrides) ·

Audit substrate (immutable event log, daily Merkle-anchor option) · LLM substrate (Claude Sonnet 4.5 + GPT-5.2, router fallbacks · rate-limit + budget alerts)

— end of brief · sources cited inline —